



# **NTC thermistors for temperature measurement**

Miniature sensors  
with bendable wires

**Series/Type:**      **B57861**  
**Date:**              March 2006

### Applications

- Heating systems
- Industrial electronics
- Automotive electronics

### Features

- Fast response
- High measuring accuracy
- Different tolerances available
- Epoxy resin encapsulation
- PTFE-insulated leads of silver-plated nickel wire, AWG 30
- UL approval (E69802)

### Options

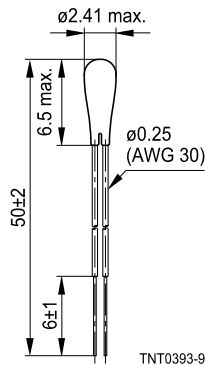
Non-standard lead lengths.

Also available with close B value tolerance.

### Delivery mode

Bulk

### Dimensional drawing



Dimensions in mm

Approx. weight 60 mg

### General technical data

|                                        |               |                  |              |      |
|----------------------------------------|---------------|------------------|--------------|------|
| Climatic category                      | (IEC 60068-1) |                  | 55/155/56    |      |
| For types with close B value tolerance | ±0.3%         |                  | 40/100/56    |      |
| Max. power                             | (at 25 °C)    | $P_{25}$         | 60           | mW   |
| Resistance tolerance                   |               | $\Delta R_R/R_R$ | ±1, ±3, ±5   | %    |
| Rated temperature                      |               | $T_R$            | 25           | °C   |
| Dissipation factor                     | (in air)      | $\delta_{th}$    | approx. 1.5  | mW/K |
| Thermal cooling time constant          | (in air)      | $\tau_c$         | approx. 15   | s    |
| Heat capacity                          |               | $C_{th}$         | approx. 22.5 | mJ/K |

## Electrical specification and ordering codes

| $R_{25}$<br>$\Omega$                                                                | No. of R/T<br>characteristic | $B_{25/100}$<br>K | Ordering code   |
|-------------------------------------------------------------------------------------|------------------------------|-------------------|-----------------|
| B value tolerance = 0.3%, resistance tolerance = $\pm 1\%$                          |                              |                   |                 |
| 5 k                                                                                 | 8016                         | $3988 \pm 0.3\%$  | B57861S0502F045 |
| 10 k                                                                                | 8016                         | $3988 \pm 0.3\%$  | B57861S0103F045 |
| 30 k                                                                                | 8018                         | $3964 \pm 0.3\%$  | B57861S0303F045 |
| B value tolerance = 1.0%, resistance tolerance = $\pm 1\%$ , $\pm 3\%$ or $\pm 5\%$ |                              |                   |                 |
| 2 k                                                                                 | 1008                         | $3560 \pm 1\%$    | B57861S0202+040 |
| 3 k                                                                                 | 8016                         | $3988 \pm 1\%$    | B57861S0302+040 |
| 5 k                                                                                 | 8016                         | $3988 \pm 1\%$    | B57861S0502+040 |
| 10 k                                                                                | 8016                         | $3988 \pm 1\%$    | B57861S0103+040 |
| 30 k                                                                                | 8018                         | $3964 \pm 1\%$    | B57861S0303+040 |
| 50 k                                                                                | 2901                         | $3760 \pm 1\%$    | B57861S0503+040 |
| 100 k                                                                               | 2014                         | $4540 \pm 1\%$    | B57861S0104+040 |

+ = Resistance tolerance

F =  $\pm 1\%$

H =  $\pm 3\%$

J =  $\pm 5\%$

## Reliability data

| Test                                  | Standard       | Test conditions                                                                           | $\Delta R_{25}/R_{25}$<br>(typical) | Remarks           |
|---------------------------------------|----------------|-------------------------------------------------------------------------------------------|-------------------------------------|-------------------|
| Storage in dry heat                   | IEC 60068-2-2  | Storage at upper category temperature<br>T: 155 °C<br>t: 1000 h                           | < 2%                                | No visible damage |
| Storage in damp heat, steady state    | IEC 60068-2-78 | Temperature of air: 40 °C<br>Relative humidity of air: 93%<br>Duration: 56 days           | < 1%                                | No visible damage |
| Rapid temperature cycling             | IEC 60068-2-14 | Lower test temperature: -55 °C<br>Upper test temperature: 155 °C<br>Number of cycles: 100 | < 1%                                | No visible damage |
| Long-term stability (empirical value) |                | Temperature: 70 °C<br>t: 10000 h                                                          | < 2%                                | No visible damage |

|         | <b>B57861S0503J040</b>                                                                                                             |                    |                    |                         |                               |                          |
|---------|------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|-------------------------|-------------------------------|--------------------------|
| R/T No. | 2901                                                                                                                               |                    |                    |                         |                               |                          |
| T (°C)  | $B_{25/100} = 3760 \text{ K}$ , $R_{25} = 50000 \text{ } \Omega$ , $T_R = 25 \text{ } ^\circ\text{C}$ , $\Delta R_R/R_R = \pm 5\%$ |                    |                    |                         |                               |                          |
|         | $R_{nomL}[\Omega]$                                                                                                                 | $R_{minL}[\Omega]$ | $R_{maxL}[\Omega]$ | $\Delta R_R/R_R[\pm\%]$ | $\Delta T[\pm^\circ\text{C}]$ | $\alpha \text{ } (\%/K)$ |
| 140.0   | 1425                                                                                                                               | 1303               | 1548               | 8.6                     | 3.6                           | 2.3                      |
| 145.0   | 1269                                                                                                                               | 1159               | 1379               | 8.7                     | 3.8                           | 2.3                      |
| 150.0   | 1132                                                                                                                               | 1032               | 1231               | 8.8                     | 3.9                           | 2.3                      |
| 155.0   | 1012                                                                                                                               | 921.7              | 1101               | 8.9                     | 4.0                           | 2.2                      |

|             | <b>B57861S0104F040</b>                                                                                                              |                    |                    |                         |                               |                          |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|-------------------------|-------------------------------|--------------------------|
| R/T No.     | 2014                                                                                                                                |                    |                    |                         |                               |                          |
| T (°C)      | $B_{25/100} = 4540 \text{ K}$ , $R_{25} = 100000 \text{ } \Omega$ , $T_R = 25 \text{ } ^\circ\text{C}$ , $\Delta R_R/R_R = \pm 1\%$ |                    |                    |                         |                               |                          |
|             | $R_{nomL}[\Omega]$                                                                                                                  | $R_{minL}[\Omega]$ | $R_{maxL}[\Omega]$ | $\Delta R_R/R_R[\pm\%]$ | $\Delta T[\pm^\circ\text{C}]$ | $\alpha \text{ } (\%/K)$ |
| –55.0       | 14200000                                                                                                                            | 13258000           | 15142000           | 6.6                     | 0.8                           | 7.8                      |
| –50.0       | 9661500                                                                                                                             | 9065600            | 10257000           | 6.2                     | 0.8                           | 7.6                      |
| –45.0       | 6656200                                                                                                                             | 6275300            | 7037000            | 5.7                     | 0.8                           | 7.3                      |
| –40.0       | 4640000                                                                                                                             | 4394300            | 4885700            | 5.3                     | 0.7                           | 7.1                      |
| –35.0       | 3270800                                                                                                                             | 3111000            | 3430600            | 4.9                     | 0.7                           | 6.9                      |
| –30.0       | 2330200                                                                                                                             | 2225500            | 2435000            | 4.5                     | 0.7                           | 6.7                      |
| –25.0       | 1677000                                                                                                                             | 1607900            | 1746100            | 4.1                     | 0.6                           | 6.5                      |
| –20.0       | 1218600                                                                                                                             | 1172800            | 1264400            | 3.8                     | 0.6                           | 6.3                      |
| –15.0       | 893700                                                                                                                              | 863230             | 924170             | 3.4                     | 0.6                           | 6.1                      |
| –10.0       | 661250                                                                                                                              | 640920             | 681590             | 3.1                     | 0.5                           | 5.9                      |
| –5.0        | 493420                                                                                                                              | 479840             | 507010             | 2.8                     | 0.5                           | 5.8                      |
| 0.0         | 371200                                                                                                                              | 362130             | 380270             | 2.4                     | 0.4                           | 5.6                      |
| 5.0         | 281450                                                                                                                              | 275410             | 287480             | 2.1                     | 0.4                           | 5.5                      |
| 10.0        | 215000                                                                                                                              | 211010             | 219000             | 1.9                     | 0.3                           | 5.3                      |
| 15.0        | 165440                                                                                                                              | 162830             | 168050             | 1.6                     | 0.3                           | 5.2                      |
| 20.0        | 128190                                                                                                                              | 126510             | 129870             | 1.3                     | 0.3                           | 5.0                      |
| <b>25.0</b> | <b>100000</b>                                                                                                                       | <b>99000</b>       | <b>101000</b>      | <b>1.0</b>              | <b>0.2</b>                    | <b>4.9</b>               |
| 30.0        | 78514                                                                                                                               | 77493              | 79536              | 1.3                     | 0.3                           | 4.8                      |
| 35.0        | 62031                                                                                                                               | 61073              | 62989              | 1.5                     | 0.3                           | 4.7                      |
| 40.0        | 49304                                                                                                                               | 48427              | 50181              | 1.8                     | 0.4                           | 4.5                      |
| 45.0        | 39417                                                                                                                               | 38626              | 40208              | 2.0                     | 0.5                           | 4.4                      |
| 50.0        | 31690                                                                                                                               | 30984              | 32396              | 2.2                     | 0.5                           | 4.3                      |
| 55.0        | 25616                                                                                                                               | 24990              | 26242              | 2.4                     | 0.6                           | 4.2                      |
| 60.0        | 20815                                                                                                                               | 20263              | 21367              | 2.6                     | 0.6                           | 4.1                      |
| 65.0        | 17000                                                                                                                               | 16515              | 17484              | 2.9                     | 0.7                           | 4.0                      |
| 70.0        | 13952                                                                                                                               | 13527              | 14377              | 3.0                     | 0.8                           | 3.9                      |
| 75.0        | 11505                                                                                                                               | 11132              | 11877              | 3.2                     | 0.8                           | 3.8                      |
| 80.0        | 9530                                                                                                                                | 9204               | 9856               | 3.4                     | 0.9                           | 3.7                      |
| 85.0        | 7930                                                                                                                                | 7644               | 8215               | 3.6                     | 1.0                           | 3.6                      |
| 90.0        | 6626                                                                                                                                | 6376               | 6876               | 3.8                     | 1.1                           | 3.5                      |

|         | <b>B57861S0104F040</b>                                                                                                   |                      |                      |                                      |         |         |
|---------|--------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|--------------------------------------|---------|---------|
| R/T No. | 2014                                                                                                                     |                      |                      |                                      |         |         |
| T (°C)  | B <sub>25/100</sub> = 4540 K, R <sub>25</sub> = 100000 Ω, T <sub>R</sub> = 25 °C, ΔR <sub>R</sub> /R <sub>R</sub> = ± 1% |                      |                      |                                      |         |         |
|         | R <sub>nom</sub> [Ω]                                                                                                     | R <sub>min</sub> [Ω] | R <sub>max</sub> [Ω] | ΔR <sub>R</sub> /R <sub>R</sub> [±%] | ΔT[±°C] | α (%/K) |
| 95.0    | 5560                                                                                                                     | 5341                 | 5779                 | 3.9                                  | 1.1     | 3.5     |
| 100.0   | 4684                                                                                                                     | 4492                 | 4877                 | 4.1                                  | 1.2     | 3.4     |
| 105.0   | 3962                                                                                                                     | 3793                 | 4131                 | 4.3                                  | 1.3     | 3.3     |
| 110.0   | 3363                                                                                                                     | 3214                 | 3512                 | 4.4                                  | 1.4     | 3.2     |
| 115.0   | 2866                                                                                                                     | 2735                 | 2997                 | 4.6                                  | 1.4     | 3.2     |
| 120.0   | 2451                                                                                                                     | 2335                 | 2566                 | 4.7                                  | 1.5     | 3.1     |
| 125.0   | 2103                                                                                                                     | 2000                 | 2205                 | 4.9                                  | 1.6     | 3.0     |
| 130.0   | 1810                                                                                                                     | 1719                 | 1901                 | 5.0                                  | 1.7     | 3.0     |
| 135.0   | 1563                                                                                                                     | 1483                 | 1644                 | 5.2                                  | 1.8     | 2.9     |
| 140.0   | 1354                                                                                                                     | 1283                 | 1426                 | 5.3                                  | 1.9     | 2.8     |
| 145.0   | 1177                                                                                                                     | 1113                 | 1241                 | 5.4                                  | 2.0     | 2.8     |
| 150.0   | 1026                                                                                                                     | 968.9                | 1083                 | 5.5                                  | 2.0     | 2.7     |
| 155.0   | 896.6                                                                                                                    | 845.7                | 947.5                | 5.7                                  | 2.1     | 2.7     |

|             | <b>B57861S0104H040</b>                                                                                                   |                      |                      |                                      |            |            |
|-------------|--------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|--------------------------------------|------------|------------|
| R/T No.     | 2014                                                                                                                     |                      |                      |                                      |            |            |
| T (°C)      | B <sub>25/100</sub> = 4540 K, R <sub>25</sub> = 100000 Ω, T <sub>R</sub> = 25 °C, ΔR <sub>R</sub> /R <sub>R</sub> = ± 3% |                      |                      |                                      |            |            |
|             | R <sub>nom</sub> [Ω]                                                                                                     | R <sub>min</sub> [Ω] | R <sub>max</sub> [Ω] | ΔR <sub>R</sub> /R <sub>R</sub> [±%] | ΔT[±°C]    | α (%/K)    |
| -55.0       | 14200000                                                                                                                 | 12974000             | 15426000             | 8.6                                  | 1.1        | 7.8        |
| -50.0       | 9661500                                                                                                                  | 8872400              | 10451000             | 8.2                                  | 1.1        | 7.6        |
| -45.0       | 6656200                                                                                                                  | 6142200              | 7170200              | 7.7                                  | 1.1        | 7.3        |
| -40.0       | 4640000                                                                                                                  | 4301500              | 4978500              | 7.3                                  | 1.0        | 7.1        |
| -35.0       | 3270800                                                                                                                  | 3045600              | 3496000              | 6.9                                  | 1.0        | 6.9        |
| -30.0       | 2330200                                                                                                                  | 2178900              | 2481600              | 6.5                                  | 1.0        | 6.7        |
| -25.0       | 1677000                                                                                                                  | 1574400              | 1779600              | 6.1                                  | 0.9        | 6.5        |
| -20.0       | 1218600                                                                                                                  | 1148400              | 1288700              | 5.8                                  | 0.9        | 6.3        |
| -15.0       | 893700                                                                                                                   | 845360               | 942050               | 5.4                                  | 0.9        | 6.1        |
| -10.0       | 661250                                                                                                                   | 627690               | 694810               | 5.1                                  | 0.9        | 5.9        |
| -5.0        | 493420                                                                                                                   | 469970               | 516880               | 4.8                                  | 0.8        | 5.8        |
| 0.0         | 371200                                                                                                                   | 354710               | 387700               | 4.4                                  | 0.8        | 5.6        |
| 5.0         | 281450                                                                                                                   | 269780               | 293110               | 4.1                                  | 0.8        | 5.5        |
| 10.0        | 215000                                                                                                                   | 206710               | 223300               | 3.9                                  | 0.7        | 5.3        |
| 15.0        | 165440                                                                                                                   | 159520               | 171360               | 3.6                                  | 0.7        | 5.2        |
| 20.0        | 128190                                                                                                                   | 123950               | 132440               | 3.3                                  | 0.7        | 5.0        |
| <b>25.0</b> | <b>100000</b>                                                                                                            | <b>97000</b>         | <b>103000</b>        | <b>3.0</b>                           | <b>0.6</b> | <b>4.9</b> |
| 30.0        | 78514                                                                                                                    | 75922                | 81106                | 3.3                                  | 0.7        | 4.8        |
| 35.0        | 62031                                                                                                                    | 59832                | 64229                | 3.5                                  | 0.8        | 4.7        |
| 40.0        | 49304                                                                                                                    | 47441                | 51168                | 3.8                                  | 0.8        | 4.5        |
| 45.0        | 39417                                                                                                                    | 37837                | 40996                | 4.0                                  | 0.9        | 4.4        |

## Cautions and warnings

### General

See "Important notes" at the end of this document.

### Storage

- Store thermistors only in original packaging. Do not open the package before storage.
- Storage conditions in original packaging: storage temperature  $-25\text{ }^{\circ}\text{C} \dots +45\text{ }^{\circ}\text{C}$ , relative humidity  $\leq 75\%$  annual mean, maximum 95%, dew precipitation is inadmissible.
- Do not store SMDs where they are exposed to heat or direct sunlight. Otherwise, the packing material may be deformed or SMDs may stick together, causing problems during mounting.
- Avoid contamination of thermistors surface during storage, handling and processing.
- Avoid storage of thermistor in harmful environments like corrosive gases (SO<sub>x</sub>, Cl etc).
- After opening the factory seals, such as polyvinyl-sealed packages, use the SMDs as soon as possible.
- Solder thermistors after shipment from EPCOS within the time specified:  
SMDs: 12 months  
Leaded components: 24 months

### Handling

- NTC thermistors must not be dropped. Chip-offs must not be caused during handling of NTCs.
- Components must not be touched with bare hands. Gloves are recommended.
- Avoid contamination of thermistor surface during handling.

### Soldering

- Use resin-type flux or non-activated flux.
- Insufficient preheating may cause ceramic cracks.
- Rapid cooling by dipping in solvent is not recommended.
- Complete removal of flux is recommended.

### Mounting

- When NTC thermistors are encapsulated with sealing material or overmolded with plastic material, the precautions given in chapter "Mounting instructions", "Sealing, potting and overmolding" must be observed.
- Electrode must not be scratched before/during/after the mounting process.
- Contacts and housings used for assembly with thermistor have to be clean before mounting.
- During operation, the thermistor's surface temperature can be very high (ICL). Ensure that adjacent components are placed at a sufficient distance from the thermistor to allow for proper cooling of the thermistors.
- Ensure that adjacent materials are designed for operation at temperatures comparable to the surface temperature of the thermistor. Be sure that surrounding parts and materials can withstand this temperature.
- Make sure that thermistors (ICLs) are adequately ventilated to avoid overheating.
- Avoid contamination of thermistor surface during processing.

## Operation

- Use thermistors only within the specified operating temperature range.
- Use thermistors only within the specified voltage and current ranges (ICLs).
- Environmental conditions must not harm the thermistors. Use thermistors only in normal atmospheric conditions.
- Contact of NTC thermistors with any liquids and solvents should be prevented. It must be ensured that no water enters the NTC thermistor (e.g. through plug terminals). For measurement purposes (checking the specified resistance vs. temperature), the component must not be immersed in water but in suitable liquids (e.g. Galden).
- Avoid dewing and condensation.
- Be sure to provide an appropriate fail-safe function to prevent secondary product damage caused by malfunction (e.g. use VDR for limitation of overvoltage condition).

## Important notes

The following applies to all products named in this publication:

1. Some parts of this publication contain **statements about the suitability of our products for certain areas of application**. These statements are based on our knowledge of typical requirements that are often placed on our products in the areas of application concerned. We nevertheless expressly point out **that such statements cannot be regarded as binding statements about the suitability of our products for a particular customer application**. As a rule, EPCOS is either unfamiliar with individual customer applications or less familiar with them than the customers themselves. For these reasons, it is always ultimately incumbent on the customer to check and decide whether an EPCOS product with the properties described in the product specification is suitable for use in a particular customer application.
2. We also point out that **in individual cases, a malfunction of passive electronic components or failure before the end of their usual service life cannot be completely ruled out in the current state of the art, even if they are operated as specified**. In customer applications requiring a very high level of operational safety and especially in customer applications in which the malfunction or failure of a passive electronic component could endanger human life or health (e.g. in accident prevention or life-saving systems), it must therefore be ensured by means of suitable design of the customer application or other action taken by the customer (e.g. installation of protective circuitry or redundancy) that no injury or damage is sustained by third parties in the event of malfunction or failure of a passive electronic component.
3. **The warnings, cautions and product-specific notes must be observed.**
4. In order to satisfy certain technical requirements, **some of the products described in this publication may contain substances subject to restrictions in certain jurisdictions (e.g. because they are classed as “hazardous”)**. Useful information on this will be found in our Material Data Sheets on the Internet ([www.epcos.com/material](http://www.epcos.com/material)). Should you have any more detailed questions, please contact our sales offices.
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